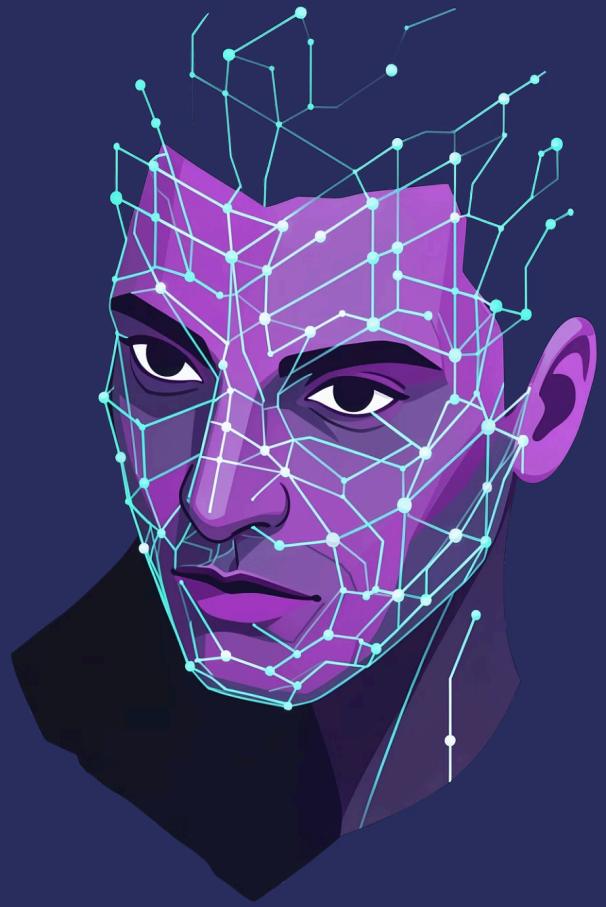


Face Expression Recognition: CNN-Based Classification

A deep learning pipeline for multi-class facial expression classification using Convolutional Neural Networks.

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Project Objective

Seven Target Classes

Happy, Fear, Disgust, Sad, Surprise, Angry, Neutral — one model to classify them all.

CNN Feature Learning

Exploit spatial hierarchies: edges, textures, shapes, and facial muscle patterns learned automatically.

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Dataset Overview

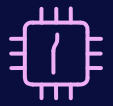
Corpus at a Glance

- **Source:** Kaggle facial-expression-dataset
- **Total images:** 35,887
- **Train / Test:** 28,709 / 7,178
- **Input format:** 48 × 48 grayscale



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Technology Stack



Environment

Google Colab with NVIDIA T4 GPU — cloud-native, no local setup required.



Framework

TensorFlow / Keras for model definition, training, and evaluation.

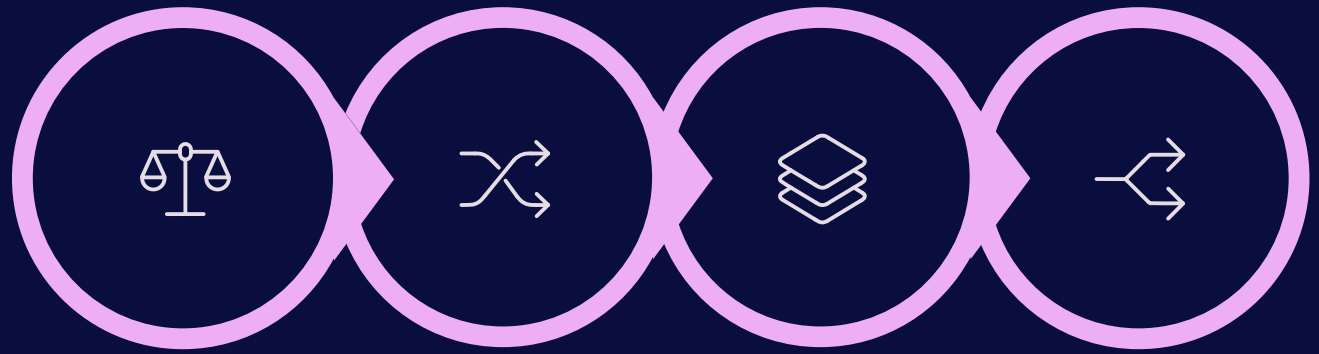


Supporting Libraries

NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn for data handling and metrics.

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Data Preprocessing Pipeline



Pipeline Details

- **Image size:** 48 × 48 grayscale, batch size 32
- **Normalization:** Pixel rescaling via 1/255
- **Augmentation:** Horizontal flip, random rotation, zoom, contrast adjustment
- **Validation split:** 80/20 using TensorFlow `validation_split`

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CNN Architecture

1

Block 1 — Feature Extraction

Conv2D(32) → BatchNorm → Conv2D(32) → MaxPool → Dropout(0.25)

2

Block 2 — Deeper Features

Conv2D(64) → BatchNorm → Conv2D(64) → MaxPool → Dropout(0.25)

3

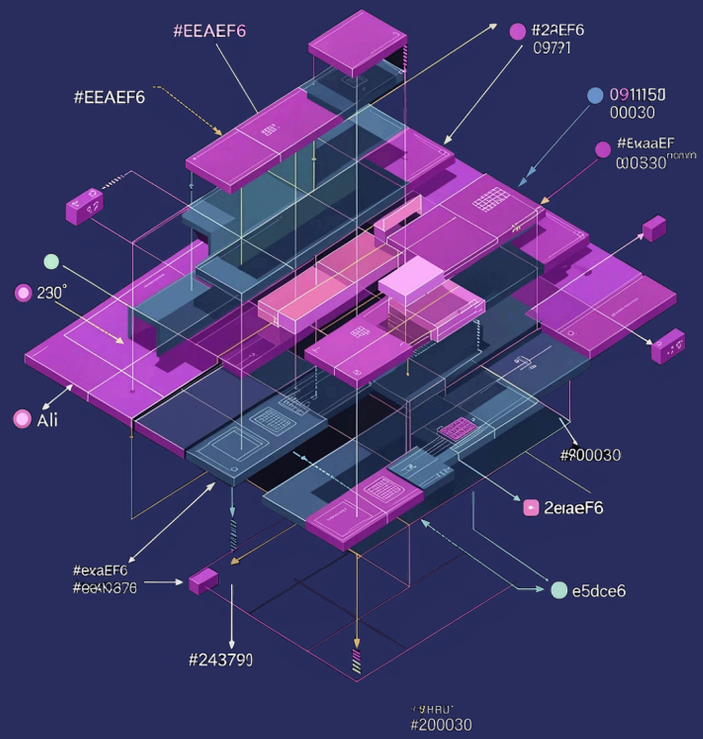
Block 3 — High-Level Patterns

Conv2D(128) → BatchNorm → Conv2D(128) → MaxPool → Dropout(0.20)

4

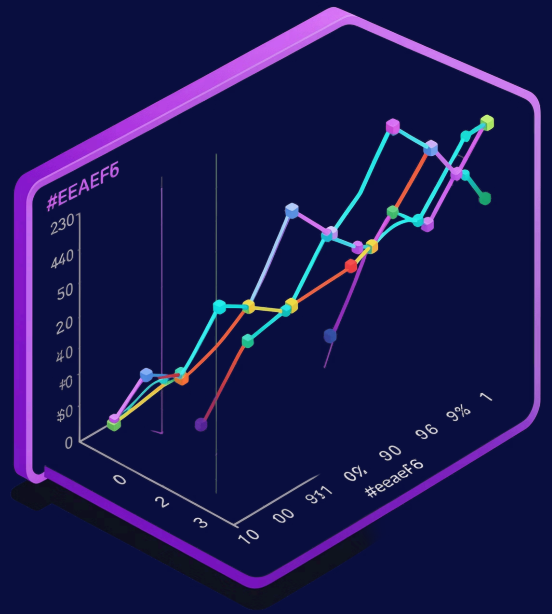
Classifier Head

Flatten → Dense(256) → BatchNorm → Dropout(0.50) → Dense(7, softmax)



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Training Configuration



Optimizer & Loss

- **Optimizer:** Adam (lr = 0.001)
- **Loss:** Sparse categorical cross-entropy
- **Epochs:** Up to 40 with early stopping (patience 7)

Callbacks

- **EarlyStopping** — prevents overfitting
- **ReduceLROnPlateau** — halves LR on plateau (factor 0.5)
- **ModelCheckpoint** — saves best weights

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Recorded Results & Critical Issue

79.42%

Validation Accuracy

Validation loss: 0.5087

2

Classes Detected

train/test folders — not 7 expressions

0

Expression Classes

Zero facial expressions actually classified

⚠ ⚠️ The dataset was loaded from the root directory. TensorFlow detected only two classes (train/test folders) — the 79.42% accuracy reflects folder classification, **not** facial-expression recognition.

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Classification Report (Current Run)

Per-Class Breakdown

Class	Precision	Recall	F1
test	0%	0%	0.00
train	79%	100%	0.88
Macro avg	40%	50%	**0.44**

Evaluated on 7,177 validation samples. Macro F1 of 0.44 exposes the incorrect class loading directly.



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Corrective Action Required

1 Fix Dataset Paths

Root cause: Paths must point to nested `train/` and `test/` subdirectories, not the dataset root. Update both `ImageDataGenerator` calls accordingly.

2 Retrain with Correct Loader

Load training data from the nested training directory and test data from the nested test directory. The model will then see all seven expression classes as intended.

3 Address Class Imbalance

Report seven-expression accuracy after retraining. Apply class weights or oversampling to handle the severe imbalance — **Disgust: 547** vs. **Happy: 8,989**.

Once paths are corrected, the true benchmark begins. The CNN architecture is sound — only the data loading needs fixing.

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